Northstar Procurement Intelligence

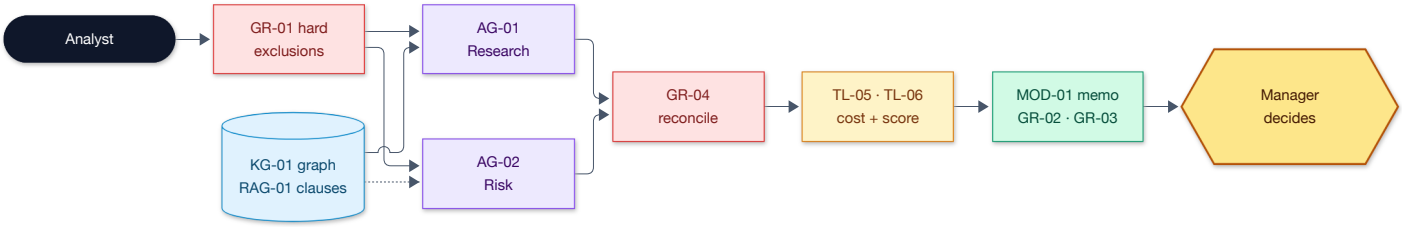
Owner: Chief Procurement Officer · **Version:** 1.0 · **Date:** 2026-11-04-17-00-00 UTC · Example 3 of 4 (illustrative numbers)

1. Problem

> **Procurement analysts** struggle to **compare suppliers on a like-for-like, compliant basis** because **quotes, contracts, certificates and ownership links sit in separate sources**, resulting in **≈ 20 h per request, 11 % cost-comparison errors and 7 % of awards later flagged for compliance or related-party issues.**

Scope: shortlist and recommendation for direct materials. **Non-goals:** issuing POs, payments, supplier contact, legal sign-off.

2. Solution: agents find evidence, the graph shows relationships, code scores, the manager decides



Component	Used for	Catalog ID
Knowledge graph + clause index + records	Ownership and certificates with provenance; contract evidence; quotes and FX	KG-01, RAG-01, REPO-01
Tools	Discover, traverse, quote, clauses, landed cost, score, approval item	TL-01…07
Models	Agents and memo (GenAI); delivery-reliability forecast (task-specific)	MOD-01, MOD-02
Workflow / agents	Fixed review order; two independent read-only agents	WF-01, WF-02, AG-01, AG-02

#	Use case	Target user	Priority
UC1	Recommend a supplier shortlist	Analyst → manager	★ riskiest
UC2	Compare quotations like for like	Analyst	High
UC3	Related-party and compliance check	Compliance reviewer	High
UC4	Incomplete or conflicting evidence	Analyst	Medium

3. Risks and guards

Risk	Guard
Cheapest but ineligible supplier recommended	`GR-01` hard exclusions run before scoring and override any score
Numbers invented in the memo	`GR-03`: every number must equal `TL-05` / `TL-06` output
Agents agree on a wrong conclusion	Independent evidence; `GR-04` turns any conflict into "needs review"

Key assumption: graph facts (ownership, certificates) are correct and current. Provenance and a 90-day staleness flag on every edge.

4. Evals

Measure	Metric	Baseline → target / threshold
Value	Analyst time per request · flagged awards	20 → ≤ 6 h · 7 → < 1 %
Quality	Shortlist accuracy · numbers = oracle · conflicts caught	≥ 90 % (exclusions 100 %) · 100 % · 5/5
Adoption	Recommendations approved without rework	≥ 75 %

Evidence so far: on 30 gold requests, 8/8 ineligible traps were excluded, 5/5 shared parents were found (text-only: 1/5) and there were 0 numeric mismatches. Time saved is still a hypothesis. **Decision asked:** the **CPO** approves an **8-week pilot** (2 categories, ≈ €35k) and names a graph data owner by **2026-12-11-17-00-00 UTC**.